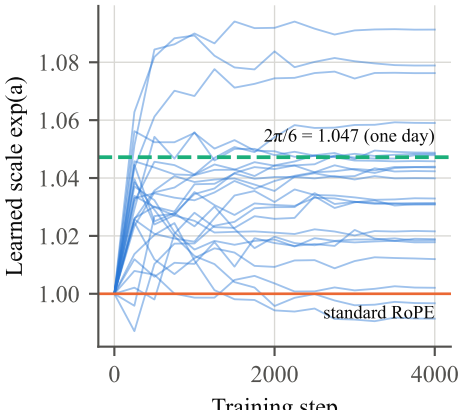
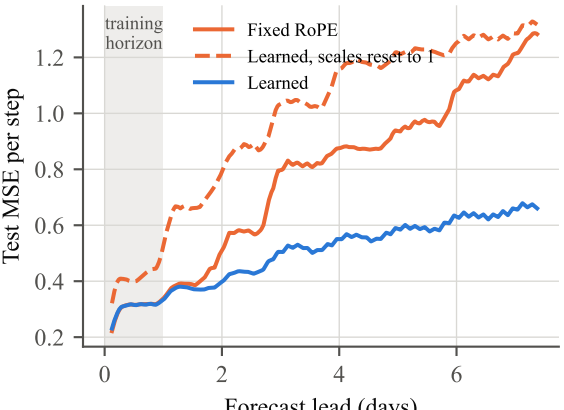

(a) 8 heads × 3 seeds, base 1e4
Learned scale exp(a)
1.08
1.06
1.04
1.02
1.00
2π/6 = 1.047 (one day)
standard RoPE
0
2000
4000
Training step


(b) Remove the learned scales at test time
training horizon
Fixed RoPE
Learned, scales reset to 1
Learned
Test MSE per step
1.2
1.0
0.8
0.6
0.4
0.2
0
2
4
6
Forecast lead (days)